

# ABHINAV BIRRU

📞 (+91) 93986 28723    ✉ [abhinavbirru@gmail.com](mailto:abhinavbirru@gmail.com)    [in abhinav-birru](https://www.linkedin.com/in/abhinav-birru)    [github.com/abhinav-birru-07](https://github.com/abhinav-birru-07)

## Education

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| <b>Indian Institute of Technology, Guwahati</b><br><i>Bachelor of Technology in Electronics and Communication Engineering</i> | <b>Nov 2020 – May 2024</b><br><i>CGPA: 8.67</i>         |
| <b>Telangana State Board of Intermediate Education</b><br><i>Senior Seocndary</i>                                             | <b>June 2018 – Mar 2020</b><br><i>Percentage: 98.5%</i> |
| <b>Jawahar Navodaya Vidyalaya, Warangal</b><br><i>Secondary</i>                                                               | <b>Mar 2018</b><br><i>Percentage: 98%</i>               |

## Experience

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| <b>AQR Capital India Services LLP</b><br><i>Software Developer</i> | <b>July 2024 - Present</b><br><i>Bengaluru, Karnataka</i> |
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### Infrastructure Modernization & Production Engineering

- Refactored a monolithic codebase into **Python 3 cloud-native modular libraries**, improving scalability and maintainability and deployment flexibility. Migrated legacy Python windows runtime to Linux-based cloud runtime, enabling more flexible updates and rollbacks and reliable production execution.
- Built and maintained **Dockerized services** to ensure reproducible execution across production environments.
- Migrated batch analytics workloads supporting portfolio and research computations from **Windows to Linux on AWS Batch**, reducing execution time and runtime failures.
- Led file storage migration from network drives to **AWS S3** with versioning and validation using **Pandas** and **NumPy**.

### Research–Production Environment Consolidation

- Owned and led consolidation of research and production environments, eliminating **100%** of redundant research jobs by redirecting workflows to stable, signed-off production data.
- Analyzed end-to-end job dependencies and data flows to decommission parallel research pipelines.
- Consolidated data ingestion and portfolio generation into a unified production-grade pipeline, cutting support effort while reducing **compute cost and operational complexity**.
- Improved research reliability by enabling analysts to work directly on **production-quality data**, reducing research–production discrepancies and increasing confidence in results.

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| <b>AQR Capital India Services LLP</b><br><i>Software Engineering Intern</i> | <b>May 2023 - July 2023</b><br><i>Bengaluru, Karnataka</i> |
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### Analytics Monitoring Dashboard (Python)

- Developed Python-based analytics library and Flask API for metrics ingestion into Elasticsearch. The library is used to capture and store metrics of running processes in Elasticsearch.
- Automated data collection from SQL and StoneBranch using time triggers in StoneBranch.
- Built Kibana dashboards for real-time process monitoring and analytics. Also developed a sophisticated dashboard using AngularJS for intuitive data visualization.
- Technologies used: **Python, Elasticsearch, AngularJS**

## Projects

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| <b>IITG-Campus-OLX</b> <a href="#"></a><br><i>Group Project, Coding Club, IITG</i> | <b>Aug.2023 - Nov 2023</b> |
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- E-Commerce Web App Development:** Designed and developed a **React JS**-based web application for buying, selling, and renting products within the IITG community.
- Firebase Integration:** Implemented **Firebase** for user authentication and utilized Firebase Storage to securely store and manage product images.
- Elasticsearch for Product Search:** Integrated **Elasticsearch** to store product details, enabling efficient and fast product search functionality for users.
- User Profiles & Product Management:** Created user profiles to facilitate quick product listing, easy editing, and seamless management of user-specific activities.

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| <b>Library Management System</b> <a href="#"></a><br><i>Personal Project</i> | <b>Jun.2022 - July 2022</b> |
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- Flask Web Application:** Designed and implemented a **Flask** and **MySQL**-based web application for efficient library book management, catering to both users and librarians.
- Role-Based Interfaces:** Developed separate, user-friendly interfaces for librarians (for book inventory management) and users (for book transactions and personalization).
- User Authentication & Features:** Implemented secure user authentication and functionalities, enabling users to add, issue, reserve books, maintain personal bookshelves, and view transaction histories.

## Technical Skills

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| <b>Languages:</b> C, C++, SQL, Python, Java, JavaScript*                                                                                               |
| <b>Tools/Frameworks:</b> AWS, JupyterHub, ReactJS*, NodeJS*, Jenkins, Docker, Flask, NumPy, Pandas                                                     |
| <b>Databases:</b> MSSQL, Elasticsearch, Firebase                                                                                                       |
| <b>Relevant Course Work:</b> Data Structures & Algorithms, Object Oriented Programming, Database Management System.<br><i>* Elementary proficiency</i> |

## Achievements

- Flipkart Grid 5.0:** Among the Top 0.5% teams shortlisted in the Online quiz out of 3,97,375 teams
- JEE Advanced 2020** Secured an All India Rank 4656 among 1,15,000 candidates.
- JEE Mains 2020** Secured an All India Rank 651 among 11,20,000 candidates.

## Leadership / Extracurricular

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| <b>Department Placement Coordinator</b><br><i>Centre For Career Development</i> <ul style="list-style-type: none"><li>Responsible for managing placement process for EEE department.</li></ul> | <b>May 2023 – July 2024</b><br><i>IIT Guwahati</i> |
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